<u>Labor Market Changes across NYC Census Tracts</u>

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Research Design

Background:

New York City has long been recognized as an economic hub for opportunities across various industries. However, the labor market and economy have shifted drastically due to the pandemic across the nation, and NYC is no stranger to this change. We've witnessed the long-term effects of these changes, including job shortages and a lack of hiring post-pandemic. Recent data from reported NY's hiring rate increased to 3.7% this past December, higher than the national average, but is still below pre-pandemic levels(Comptroller NYC 2026). NYC Comptroller also reported a shortage of job openings across the state, with reported layoffs that fell 1.3%. Based on this observation, it sparked my interest in analyzing how the labor market and employment have changed across NYC.

Research Questions:

Across NYC census tracts, how has employment and the labor market shifted before and after the pandemic? How have specific industries been impacted by these changes? Has employment increased or decreased post-pandemic?

Hypothesis:

H0: There is no relationship between the labor force and employment changes post-pandemic across NYC I census tracts from 2019 to 2023. There is no relationship between employment change and industry impact post-pandemic across NYC census tracts from 2019 to 2023.

HA: There is a relationship between the labor force and employment changes across NYC census tracts from 2019 to 2023. There is a relationship between labor force change and industry impact post-pandemic across NYC census tracts from 2019 to 2023

Application of Statistical Tools

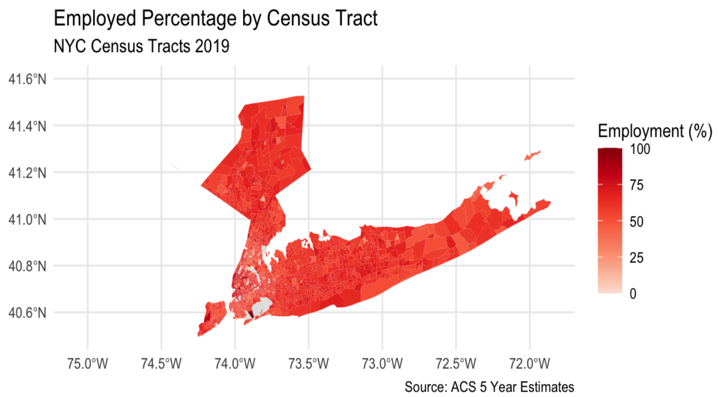
Data Sources and Pipeline:

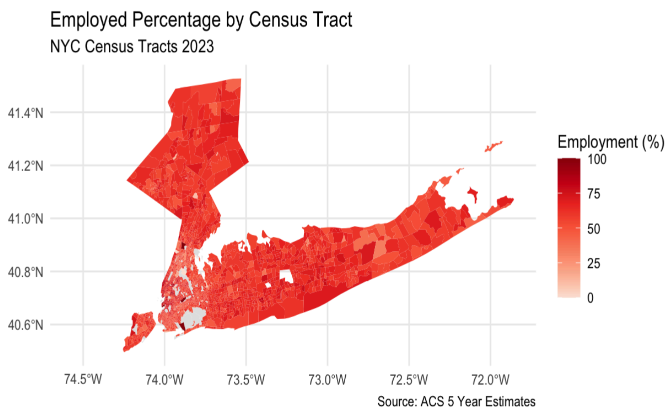
- **Data Source: Census API data: American Census Survey NYC census tract from 2019 to 2023 and pulled variables of interest: employment, labor force, unemployment, occupation, industries: arts, information, manufacturing**

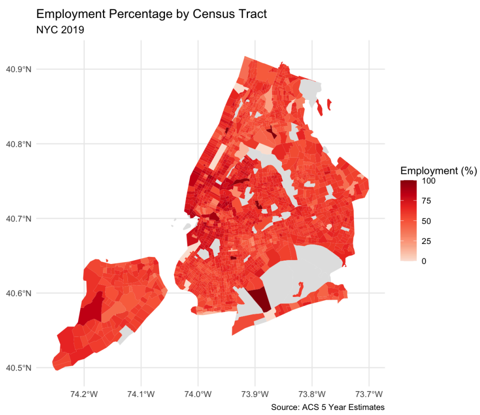
Data Cleaning: After pulling the data, I cleaned the data to more appropriate names and pulled NYC census tract data for 2019 and 2023. I transformed the data and filtered to ensure it was pulling the correct geographical data. I created several preliminary Exploratory

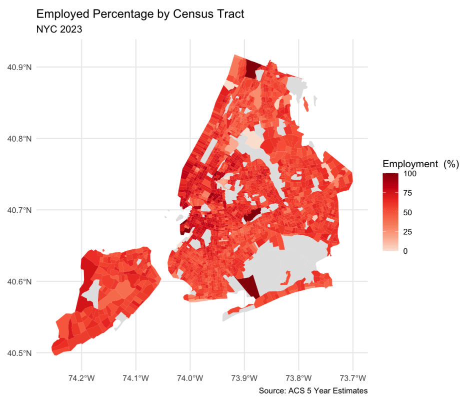
Data Analysis visualizations with the separate data sets. I wanted to analyze employment percentage across the NYC census tracts. I created Spatial Maps of how the employment percentage was distributed, filtering with a close-up of New York, and observed census tracts with specific counties within the boroughs. I wanted to see if there were any overly noticeable differences in color grading. Then I created two scatter plots that observed employment and unemployment percentages across the census tracts. Finally, I created histograms about how unemployment counts were across the population.

Exploratory Data Visualizations:
Spatial Graphs:

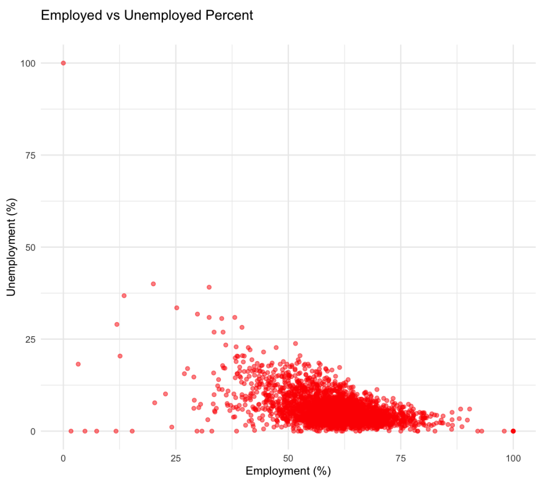


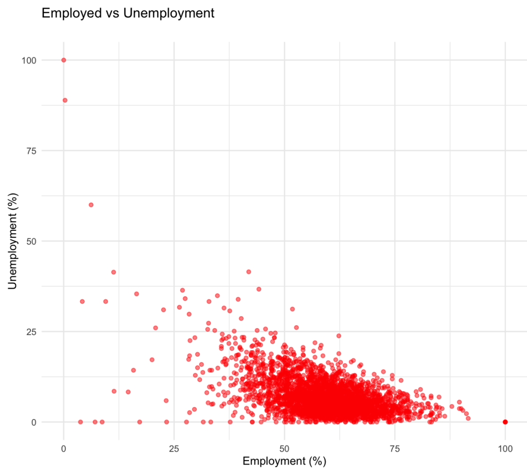




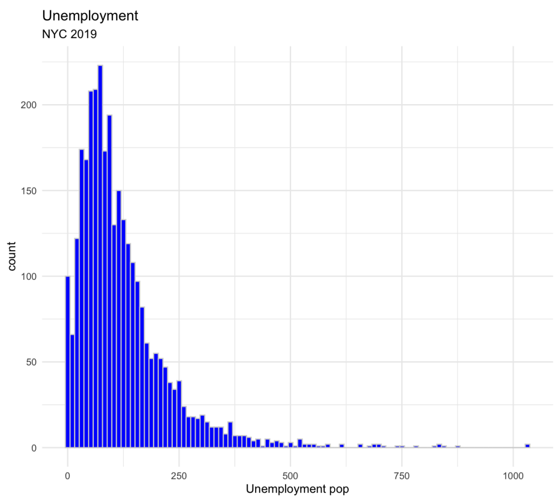


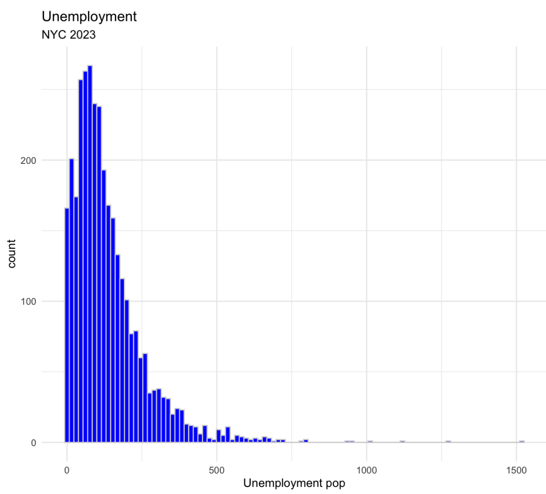
Scatterplots:





Histograms:

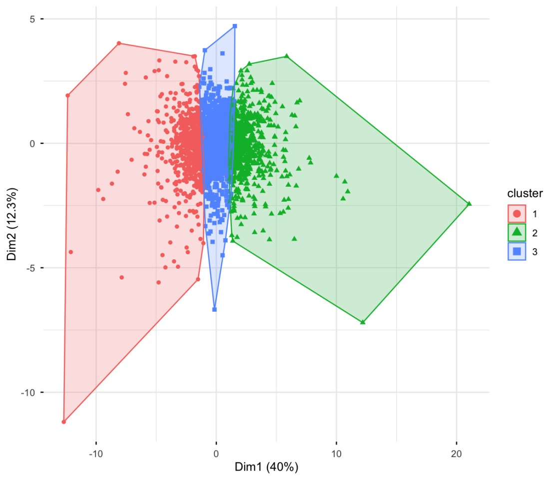




Geographical Crosswalking:

Before I could merge the data, I had to ensure the conversion of census tracts from 2019 to 2023 census tracts. Census tracts from one year might not necessarily be the same in the next year. I downloaded the IPUMS NHGIS file for 2010 to 2020 census tracts. Throughout the cleaning, I had to multiply the weights in the file by my variables of interest to redefine the boundaries of the data. I needed to convert column types, match IDs, and remove any null values. Then, I was able to join the two sets. After creating the merged dataset, I created new columns that calculated the difference between years. For example, calculating the difference between employment changes from 2023 to 2019. This was what I wanted to use to analyze further.

Clustering Analysis:



I chose to analyze a cluster analysis as a dimensionality reduction method. I pulled the new variables of interest that I mutated with the merged data such as employment change and labor force change. After scaling and removing null values, I computed the cluster visualization. Based on the analysis, the clustered groups are extremely close, but there is no visible overlap between the groups.Cluster 1 and 2 has higher averages for the variables of interest compared to Cluster 3. The cluster analysis observes that the grouping for post-pandemic changes that Cluster 3 has a more selective environment

Model Analysis:

For my model analysis, I decided to use linear regression to analyze my research hypothesis. I chose linear regression for a continuous outcome based on predictor variables.

Linear Regression Models

	Coefficients	t value	Pr(>|t|)	p value	RSE
Model 1: Employment change	Labor force change	-13.32	<2e-16	< 2.2e-16	93.39
Model 2: Arts change	Labor force c. employment c.	-12.355	<2e-16	<2.2e-16	101.7
Model 3: Information change	Labor force c. employment c.	-2.507	0.0122	<2.2e-16	56.15

	R squared	Adjusted R Squared	F-stastic
Model 1: Employment change	0.9258	0.9258	4.14e+04
Model 2: Arts change	0.1465	0.146	284.8
Model 3: Information change	0.05629	0.5573	98.96

This table summarizes some of the goodness of fit metrics for each of the models. The first model, I wanted to observe how employment changes when the labor market changes. The R-squared was 93%, which analyzes that 93% of the variance is explained by these changes. The goodness of fit metrics analyze a positive relationship between labor force changes and employment changes Model 2 was observing the change in the arts industry when the labor force and employment changed. Only 15% of the variance was explained by this change. Model 2 has a very weak relationship between changes in labor force and employment to changes in the art industry. Model 3 was observing the changes in the information industry when there were changes in the labor force and employment. Only 6% of the variance was explained by these changes. Model 3 has little to no relationship from changes in labor force and employment to the information industry.

Results/Interpretation:

The models suggest that there have been changes in the labor force and employment across NYC census tracts after the pandemic. However, the relationships are not as strong when focused on specific industries such as arts and information. The results may suggest that other factors may impact the variance more than just changes in the labor force and employment. There is a significantly weaker relationship between employment changes in the arts and information industry. The p-values are small and are statistically significant.

Evaluation of assumptions:

The assumptions of linear models are the normality of errors, homoscedasticity of errors, and the absence of overly influential outliers. The data does not meet these assumptions because the data is skewed due to outliers and the normality of the errors.

Limitations:

However, there are limitations to the data. It is at the census tract level, but the individual level would be more accurate. The data is considering variables such as income/class level and education. This information would help with the accuracy of the data. For future direction**,** Pulling other data sets to merge that observe other variables that could affect individuals on the labor market would be the next steps for better analysis.

Documentation & Reproducibility:

- **Github README goes through the keys and files needed to reproduce the R scripts such as how to request API key, IPUMS NHGIS, geographical crosswalking, etc:** https://github.com/ashstephh/Data-Analysis-Final-Project

References:

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https://github.com/johnlauermann/student-resources/blob/main/info-640-data-analysis/Lab_09.R